The Graph Laplacian at Work — Exercises

Companion to G. Oliva, *The Graph Laplacian at Work: One Least-Squares Problem Behind Consensus, Graph Drawing, Clustering and Decision Making*

Section numbers refer to the paper. Exercises marked [code] use the notebooks in `notebooks/` and the functions in `code/`. Unless stated otherwise, graphs are undirected and connected, $L = BWB^\top$, and eigenvalues are ordered increasingly.

Exercise 1 (Section 2). *Orientation does not matter, but the data must follow it*

(a) Show that reversing the orientation of edge h multiplies the h-th column of B by -1 and leaves L unchanged.

(b) In Example 2.4, orient the third edge as $e_3 = (v_1, v_3)$ instead of (v_3, v_1). Write the new B, and the edge vectors that describe the *same measurements* as $\boldsymbol{b} = (1, 1, -2, 5)^\top$ and $\boldsymbol{b} = (1, 1, 1, 5)^\top$.

(c) Which vector now spans $\ker(B)$? Check that the consistency test of Theorem 2.3(iii) gives the same verdicts as before.

Exercise 2 (Sections 2.3 and 2.5). *Four families, and a graph on which $\theta = 1/\max_i d_i$ fails*

Consider the path P_n, the cycle C_n, the star S_n (one centre and $n-1$ leaves) and the complete graph K_n, all unweighted.

(a) Find $\lambda_2(L)$ and $\lambda_n(L)$ for each family. (For C_n, try $x_k = \mathrm{e}^{2\pi \mathrm{i} jk/n}$; for P_n, $x_k = \cos\big(\pi j(k - \tfrac{1}{2})/n\big)$.)

(b) Compute the rate $\rho(\theta)$ of the discrete consensus iteration with $\theta = 1/\max_i d_i$, and evaluate it for $n = 10$.

(c) For which family is the boundary value $\theta = 1/\max_i d_i$ *not* admissible (Remark 2.6)? Describe what the iteration does from the alternating initial condition $x_k(0) = (-1)^k$.

Exercise 3 (Section 3.3). *The best constant step*

(a) Show that $\rho(\theta) = \max\{|1 - \theta\lambda_2|, |1 - \theta\lambda_n|\}$ is minimized by $\theta^\star = 2/(\lambda_2 + \lambda_n)$, and that $\rho(\theta^\star) = (\kappa - 1)/(\kappa + 1)$ with $\kappa = \lambda_n/\lambda_2$.

(b) [code] On the sensor network of Section 3, compare the number of iterations needed to reduce the error by 10^3 with $\theta = 1/\max_i d_i$ and with $\theta^\star$.

(c) Why can the agents not simply use $\theta^\star$? Suggest two ways out.

Exercise 4 (Section 4.2). *The anchored layout is a random walk*

Let $\mathcal{A} \neq \emptyset$ be the anchors and $\mathcal{F}$ the free nodes of a connected graph, and let $H = -L_{\mathcal{F}\mathcal{F}}^{-1} L_{\mathcal{F}\mathcal{A}}$, so that the anchored layout of Section 4.2 is $P_{\mathcal{F}} = H P_{\mathcal{A}}$ (for any anchors; it is Tutte's embedding when G is planar and 3-connected and the anchors are the outer face placed on a convex polygon).

(a) Show that H is entrywise nonnegative and that its rows sum to one. (Hint: write $L_{\mathcal{F}\mathcal{F}} = D_{\mathcal{F}}(I - D_{\mathcal{F}}^{-1} A_{\mathcal{F}\mathcal{F}})$ and use a Neumann series; for the row sums, use $L\mathbf{1}_n = \mathbf{0}_n$.)

(b) Deduce that every free node is drawn inside the convex hull of the anchors.

(c) Consider the random walk that moves from v_i to a neighbour v_j with probability w_{ij}/d_i. Show that H_{ia} is the probability that the walk started at v_i reaches the anchor set for the first time at v_a.

(d) [code] Check (c) by simulation on the Delaunay example.

Exercise 5 (Sections 4.2 and 4.4). *How many leaders?*

(a) Use Cauchy's interlacing theorem to show that with k anchors $\lambda_{\min}(L_{\mathcal{FF}}) \leq \lambda_{k+1}(L)$; in particular a single leader can only slow the network down.

(b) [code] On the Delaunay example, anchor the first k vertices of the convex hull (in the order returned by the code). How many are needed for the anchored consensus to beat the free one, i.e. for $\lambda_{\min}(L_{\mathcal{FF}}) > \lambda_2(L)$?

(c) [code] Choose the anchors greedily, one at a time, among *all* nodes. How many are needed now? Comment on the difference between a good anchor set for drawing and a good anchor set for speed.

Exercise 6 (Section 5.2). *Normalized cuts of haemoglobin*

(a) [code] For the *dimer* partition $A = \{\alpha_1, \beta_1\}$ (chains A, B) versus $\bar{A} = \{\alpha_2, \beta_2\}$ (chains C, D), build the two-valued vector $\boldsymbol{x}$ of Section 5.2 and verify the three identities $\boldsymbol{x}^\top D\mathbf{1} = 0$, $\boldsymbol{x}^\top D\boldsymbol{x} = \mathrm{vol}(V)$, $\boldsymbol{x}^\top L\boldsymbol{x}/\boldsymbol{x}^\top D\boldsymbol{x} = \mathrm{Ncut}(A, \bar{A})$.

(b) [code] Compute the normalized cut of the dimer partition, of the α/β partition (A, C versus B, D) and of the partition obtained by thresholding the second eigenvector of $L\boldsymbol{v} = \mu D\boldsymbol{v}$ at zero. Which is the smallest?

(c) Cheeger's inequality bounds the conductance $h_G = \min_S \mathrm{cut}(S, \bar{S})/\min\{\mathrm{vol}(S), \mathrm{vol}(\bar{S})\}$ of the *best* cut, $\mu_2/2 \leq h_G \leq \sqrt{2\mu_2}$, with μ_2 the second eigenvalue of $L\boldsymbol{v} = \mu D\boldsymbol{v}$. Compute the conductance of the three cuts of (b) and position them with respect to this bracket; then sweep a threshold through the second eigenvector and report the best sweep cut. What do (b) and the value of μ_2 say about the quality of the spectral relaxation?

Exercise 7 (Sections 5.3 and 5.4). *When does the eigengap work?*

[code] Rebuild the haemoglobin contact network with the upper cut-off $6, 6.5, \ldots, 10$ Å (the function `contact_network` in `notebooks/03_clustering.ipynb`).

(a) For which cut-offs does the graph have isolated residues? What does that do to L_{norm} and to Theorem 2.2(ii)?

(b) For the others, record $k^\star$ over the same prespecified coarse range $k = 2, \ldots, 9$ as in the paper, and the relative margin between the largest and the second largest gap in that range.

(c) Fix $k = 4$ and record the agreement with the chains. Which is more robust, the choice of k or the partition for a given k?

Exercise 8 (Section 6). *Several decision makers: $W \neq I$*

Pair $h = (v_i, v_j)$ is judged by m_h experts; expert e reports $\ln S_{ij}^{(e)} = r_i - r_j + \eta_h^{(e)}$, with independent $\eta_h^{(e)} \sim \mathcal{N}(0, \sigma^2)$, and b_h is the mean of the m_h log-judgements.

(a) Show that the maximum-likelihood estimate of $\boldsymbol{r}$ solves Problem 1 with the weight matrix

$$W = \operatorname{diag}(m_1, \dots, m_m).$$

(b) Write the distributed iteration that agent i runs, and an admissible step size.

(c) [code] On the small world of Section 6, with m_h uniform in $\{1, \dots, 5\}$ and $\sigma = 1$, compare $W = I$ and $W = \operatorname{diag}(m_h)$.

Exercise 9 (Sections 6 and 7.4). *Variance is effective resistance*

Let $W = I$, $\boldsymbol{b} = B^\top \boldsymbol{r} + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_m)$ and $\mathbf{1}^\top \boldsymbol{r} = 0$, and let $\hat{\boldsymbol{r}} = L^\dagger B \boldsymbol{b}$.

(a) Show that $\hat{\boldsymbol{r}}$ is unbiased and that $\operatorname{Cov}(\hat{\boldsymbol{r}}) = \sigma^2 L^\dagger$. Deduce $\operatorname{Var}(\hat{r}_i - \hat{r}_j) = \sigma^2 R_{ij}$, where $R_{ij} = (\boldsymbol{e}_i - \boldsymbol{e}_j)^\top L^\dagger (\boldsymbol{e}_i - \boldsymbol{e}_j)$ is the effective resistance between v_i and v_j when every edge is a unit resistor.

(b) Show that $R_{ij} \leq 1$ for adjacent nodes, with equality if and only if the edge is a bridge. Interpret: when is the least-squares estimate of $r_i - r_j$ better than the direct judgement b_{ij}?

(c) [code] On the small world of Section 6, find the range of R_{ij} over the edges, the worst-estimated pair, and verify (a) by Monte Carlo.

Exercise 10 (Section 3.4). *Directed consensus*

Take the digraph with arcs $v_1 \to v_2$, $v_2 \to v_3$, $v_3 \to v_4$, $v_4 \to v_1$, $v_1 \to v_3$, with the convention $A_{ij} = 1$ for the arc $v_i \to v_j$ and the dynamics $\dot{x}_i = \sum_j A_{ij}(x_j - x_i)$.

(a) Write L, check that $L\mathbf{1} = \mathbf{0}$, and decide whether the graph is balanced.

(b) Compute the left null vector $\boldsymbol{\gamma}$ with $\boldsymbol{\gamma}^\top \mathbf{1} = 1$ and the limit from $\boldsymbol{x}(0) = \boldsymbol{e}_1$. Is it the average?

(c) Compute the spectrum of L. What sets the rate of convergence now?

(d) Add one arc so that the graph becomes balanced, and check that the limit becomes the average.